

WEEK 3

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Dataset: BCI Competition 2008 – Graz data set A

- 9 subjects, 22 Channels, 1001 samples (250 Hz, 4 seconds of data)
- 576 trials (12 trials per class, 4 classes per run, 6 runs, 2 sessions)
- Files separated into data, labels, and metadata for each subject
- Data saved per subject (trials x channels x samples)
- **Preprocessing:**
 - Used MOABB to load the data
 - Subject data for training/validation/testing merged together
 - 6 subjects used for training
 - 2 subjects used for validation
 - 1 subject used for validation
 - Ideally will rerun but using K-fold, to avoid bias

WEEK 3

- **Simple NN model:**

- Created using Pytorch
- 3 Blocks of 2D convolutions and MaxPooling
- Worst performing accuracy, only slightly higher than chance (25%)

```
Test - [0.3142361111111111,
```

- **Imported EEGNet Model:**

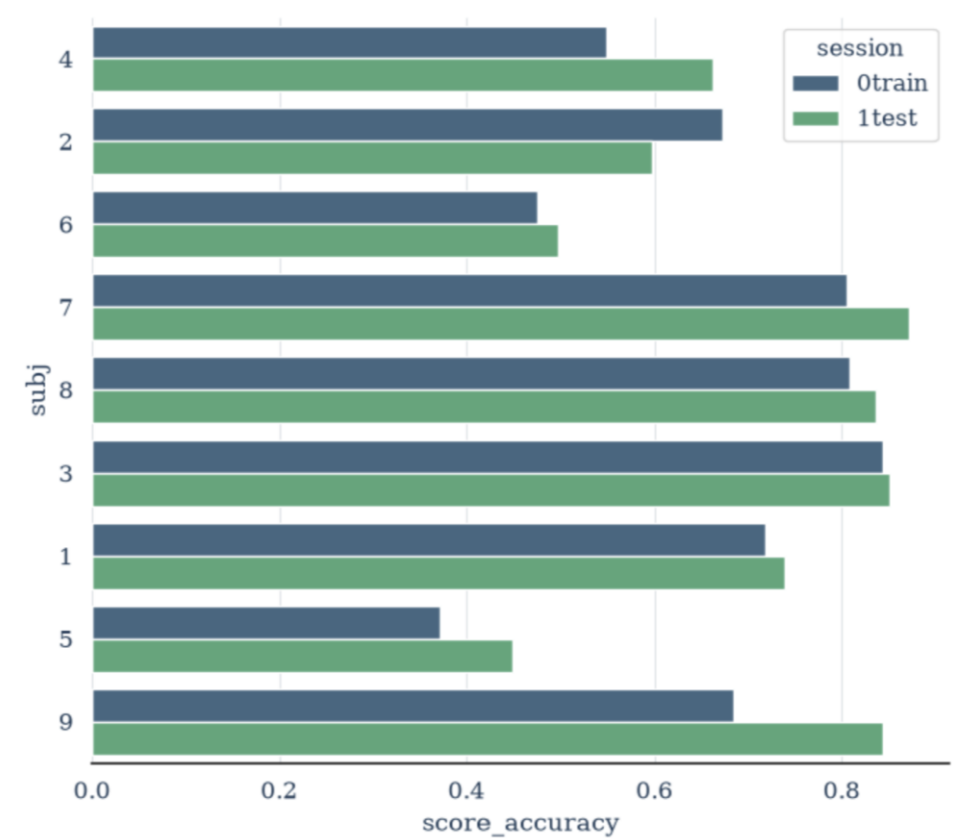
- Imported from GitHub
- Written in Keras
- Training took the longest

```
Classification accuracy: 0.513889
```

WEEK 3

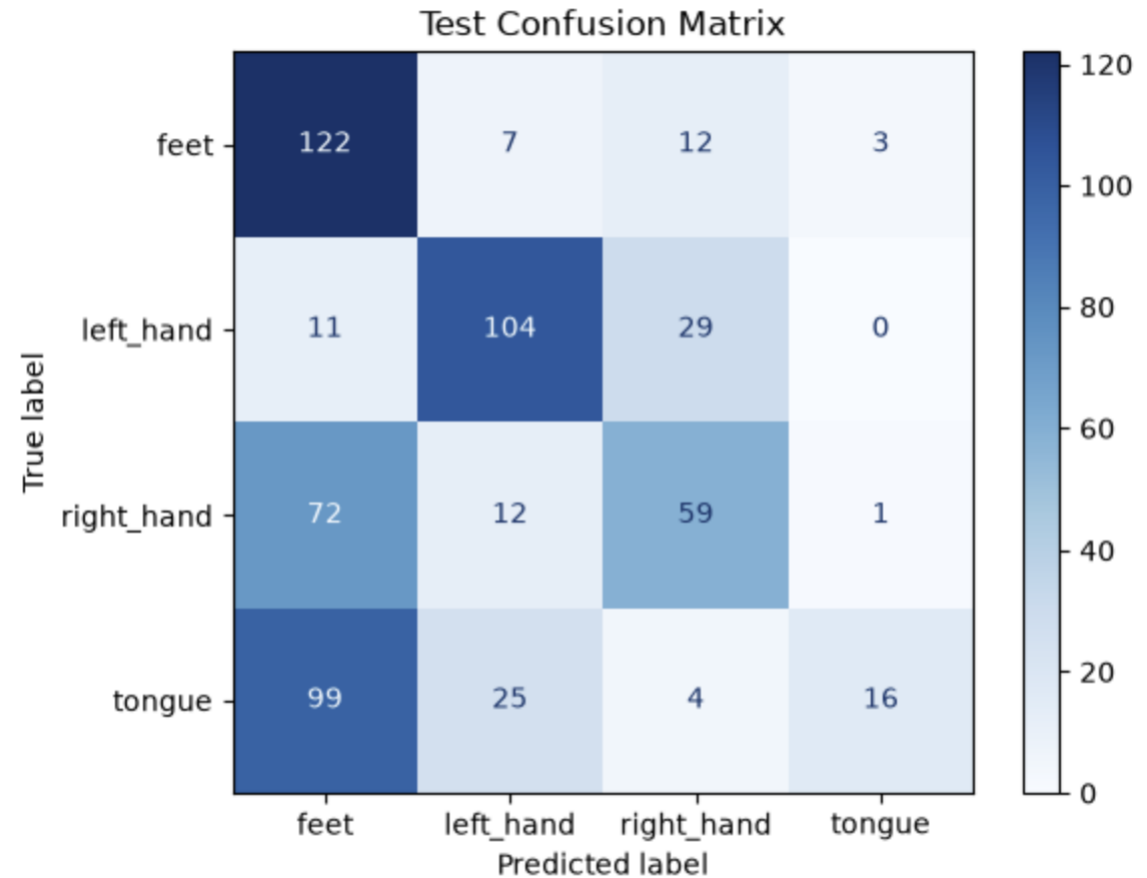
- **Simple CSP + LDA model:**
 - Mostly used for data exploration
 - Created using the MOABB library
 - “Best performance and fastest training”
 - NOTE: only model that used K-fold

	Accuracy	Balanced Accuracy	score_f1_macro
session			
0train	0.658950	0.659259	0.655046
1test	0.705209	0.705370	0.702406



WEEK 3

- EEGNet converted to Pytorch
 - Similar accuracy to the Keras model
 - However, significantly faster



WEEK 3

- Challenges:
 - **Understanding and manipulating the data**
 - Used the MOABB website, guides, and tutorials
 - **Significant difference in accuracy when converting Keras to Pytorch:**
 - Comparing the default values showed difference in multiple functions
 - Difference in layer order requirements caused multiple issues
 - **Running the original EEGNet model needed a new environment and managing the requirements**

WEEK 4

WEEK 4

Baseline: CSP + LDA model

- 70.5% accuracy

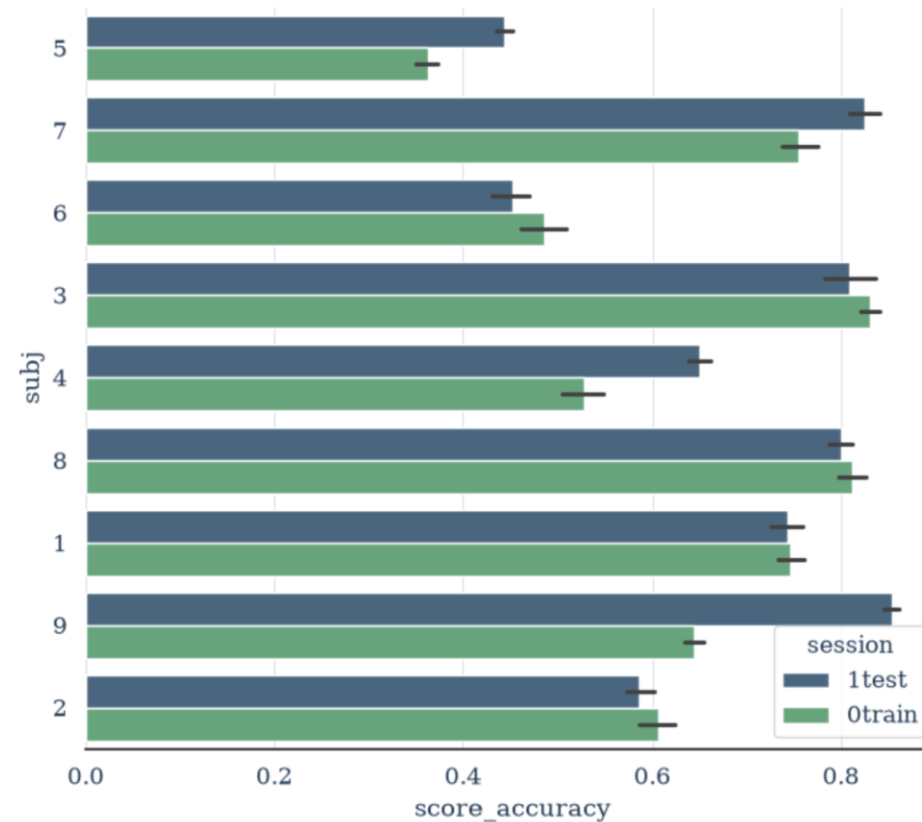
Experiments:

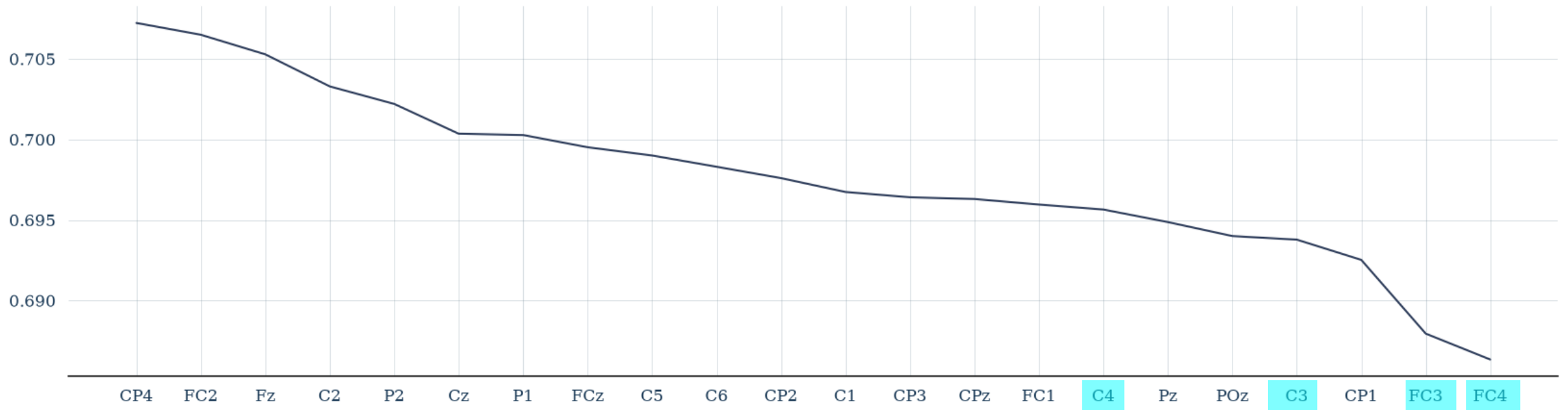
- Preprocessing on both train and test:
 - Multiple values tested, tests averaged

WEEK 4

Robustness to missing data:

- 4 missing channels selected at random, repeated 10 times and averaged
- Test accuracy: 68.5%



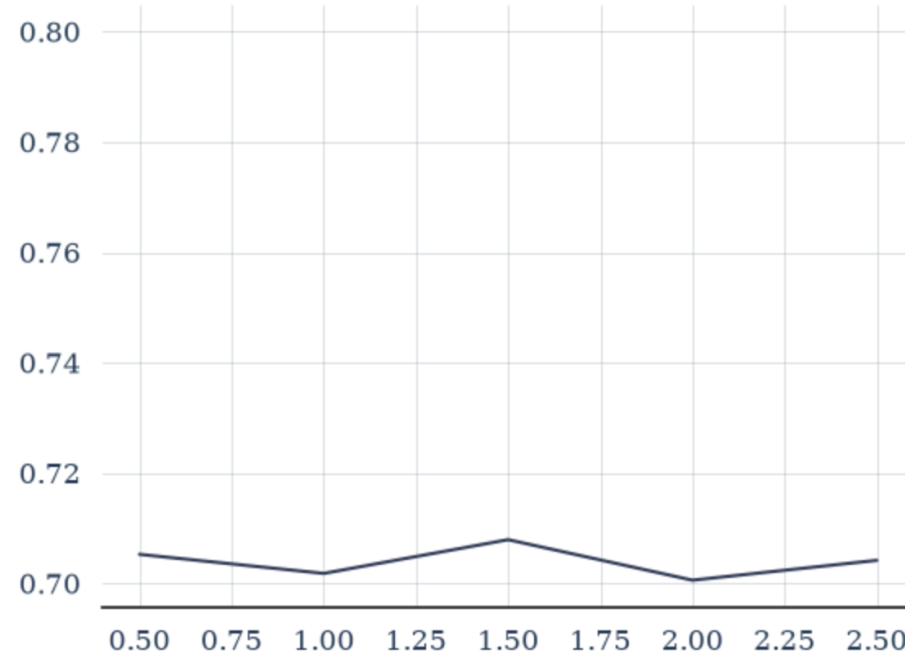


Fronto-Central: Motor planning , cognitive control, and executive function

WEEK 4

Robustness to scaling:

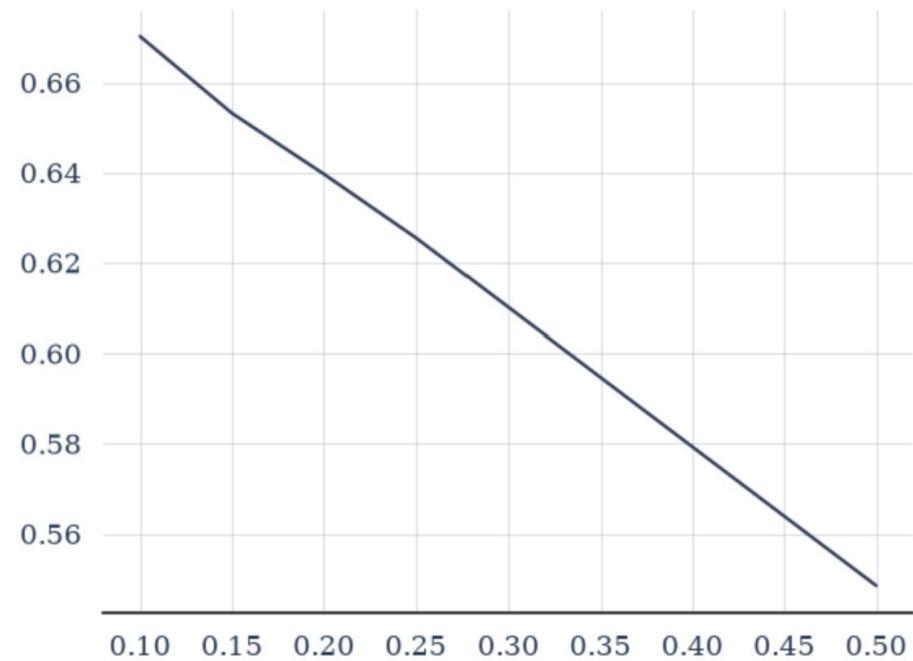
- Scaled on 4 Random channels, twice. Results were averaged
- Scaled on 5 factors = [0.5, 1, 1.5, 2, 2.5, 3, 4, 5]
- 70.5% accuracy



WEEK 4

Robustness to noise:

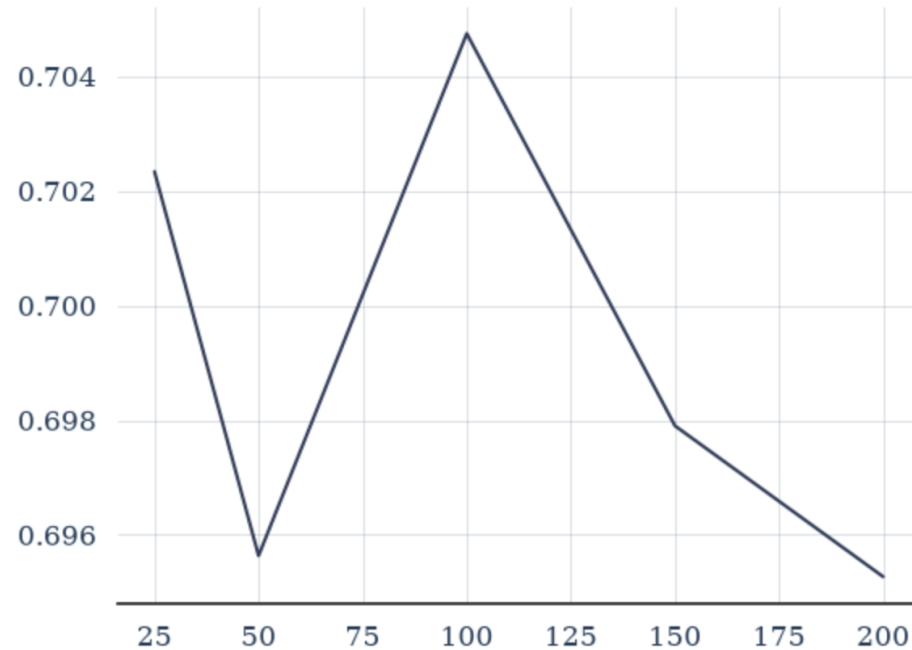
- Added noise to all channels, 5 times. Results were averaged
- Levels= 0.02-0.5
- 62.7% accuracy



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Robustness to missing durations:

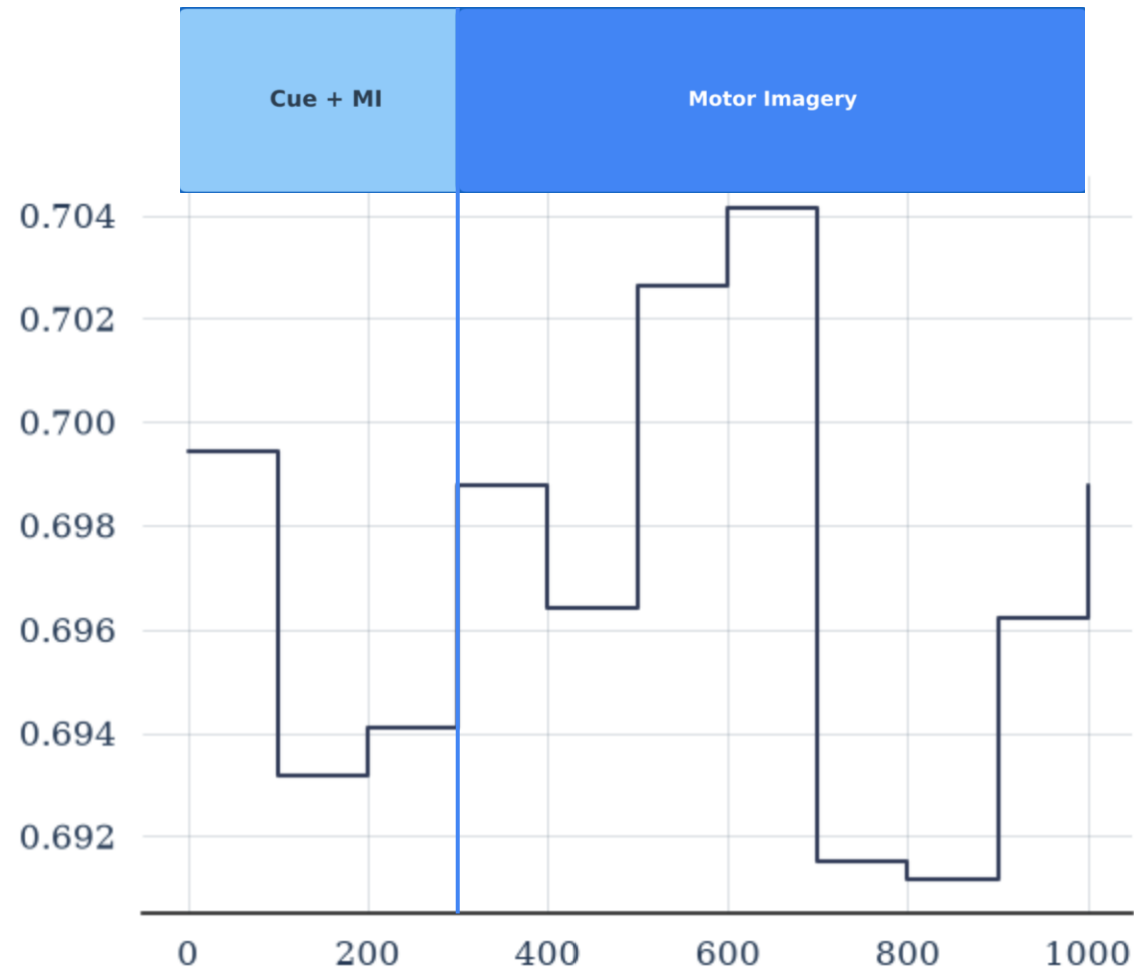
- Removed durations from a random channel, repeated twice. Results averaged
- durations=[25, 50, 100, 150, 200]
- 69.9% accuracy



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Robustness to missing durations:

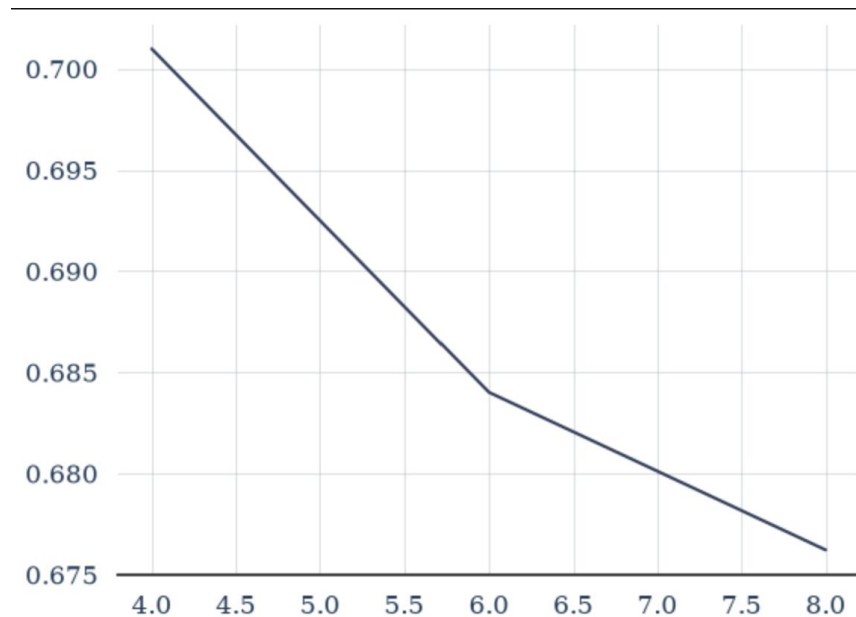
- Removed 200 durations from FC3 and FC4



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Robustness to missing frequencies:

- Removed different frequencies from bandwidths 4-8, repeated thrice. Results averaged
- Bandwidths=[4,6,8]
- 68.7% accuracy



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Summary

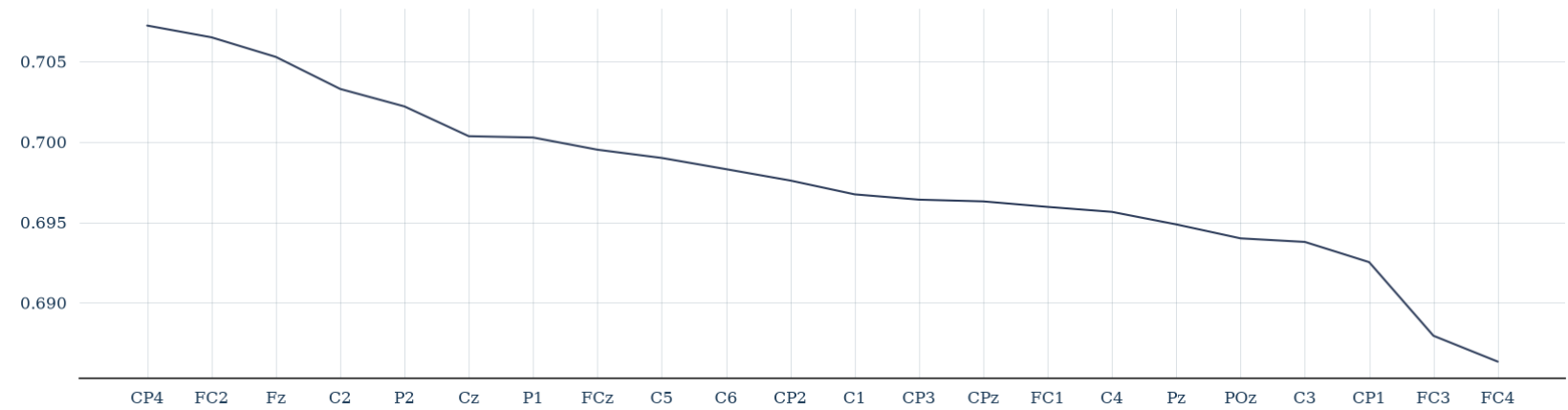
	Baseline	Scaling	Missing durations	Missing frequencies	Missing data	Noise
Accuracy	70.5%	70.5%	69.9%	68.7%	68.5%	62.7%

WEEK 5

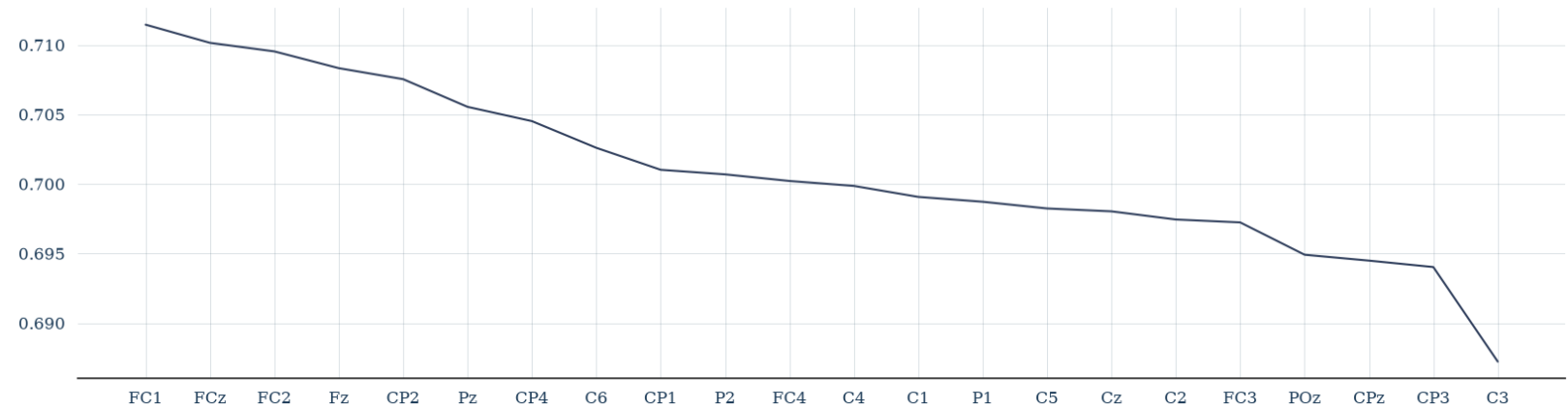
WEEK 5

Missing data:

Both preprocessed:



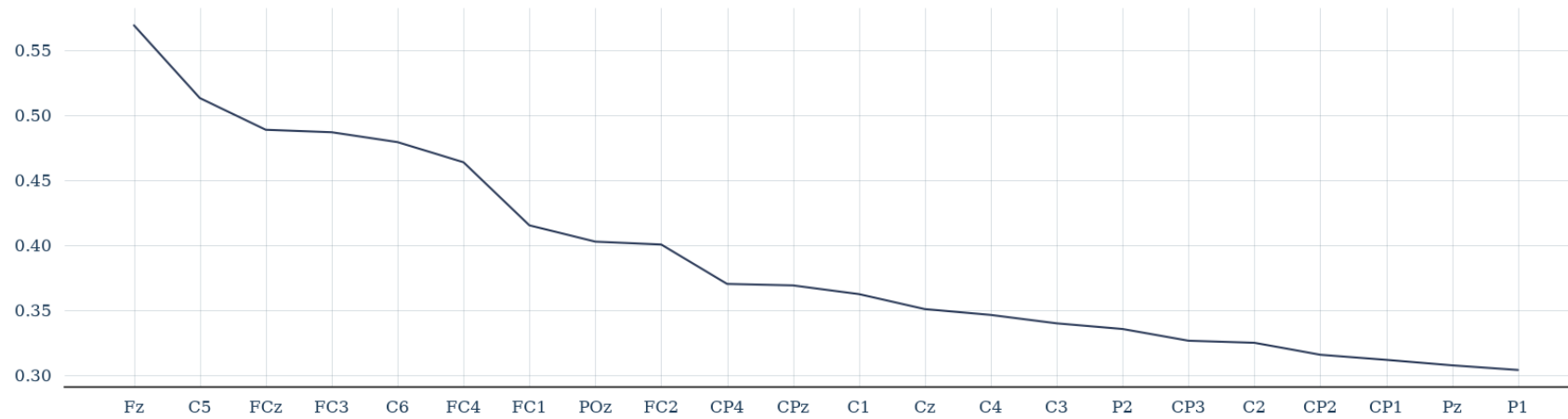
Train only preprocessed:



WEEK 5

Missing data:

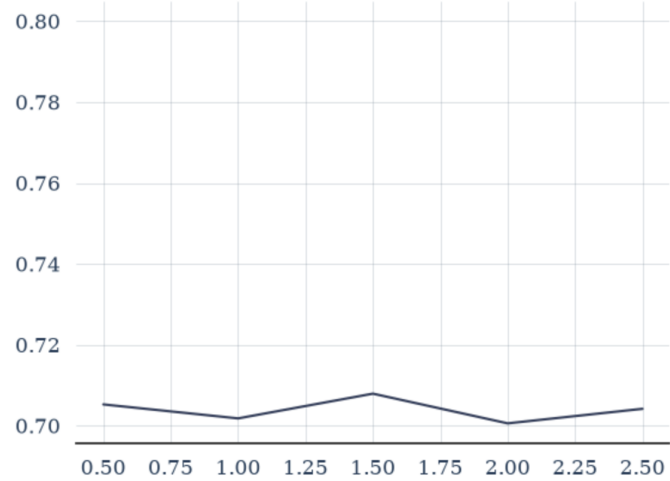
Test only preprocessed:



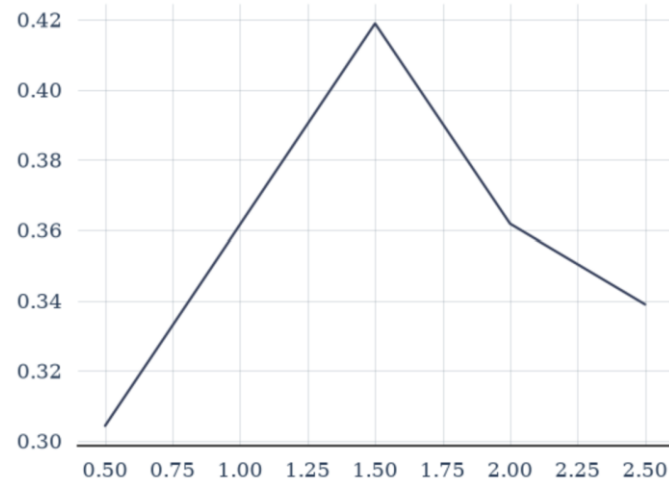
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Scaling:

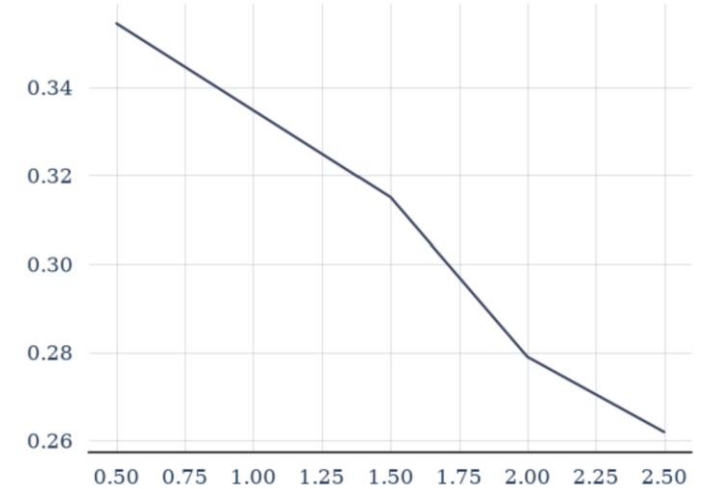
Both preprocessed



Train preprocessed



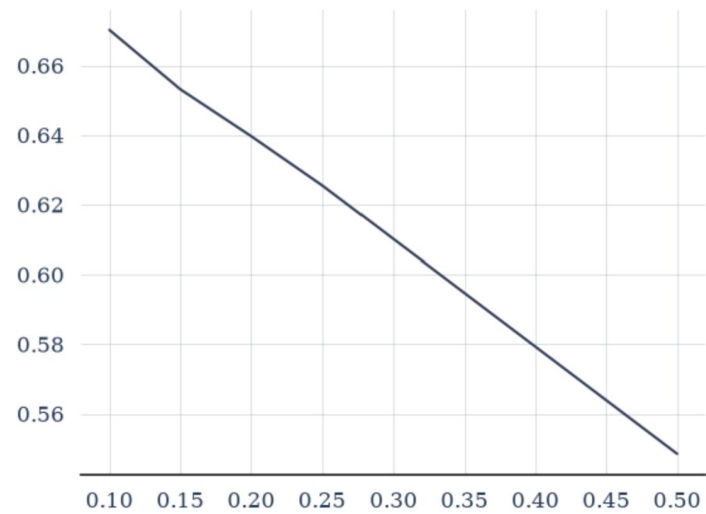
Test preprocessed



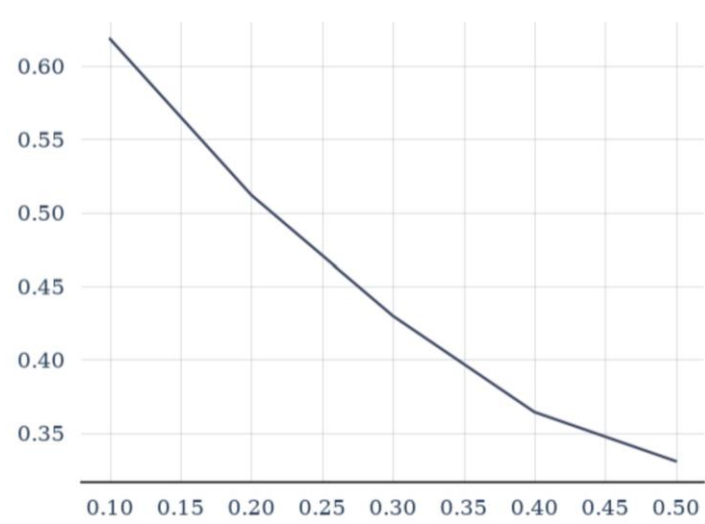
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Noise:

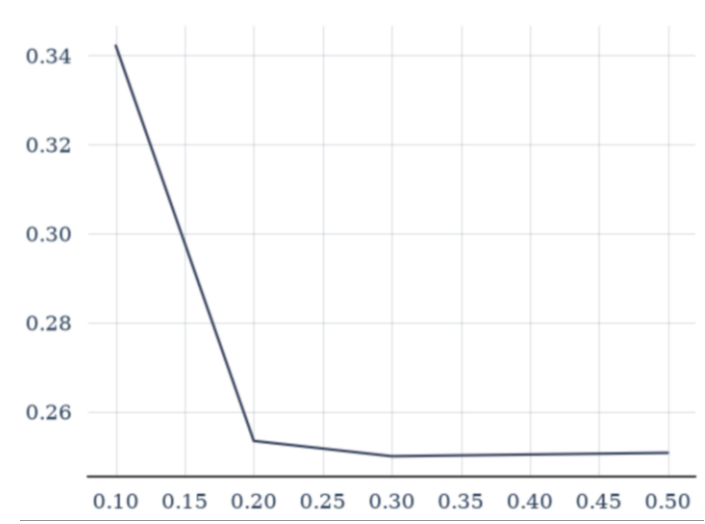
Both preprocessed



Train preprocessed

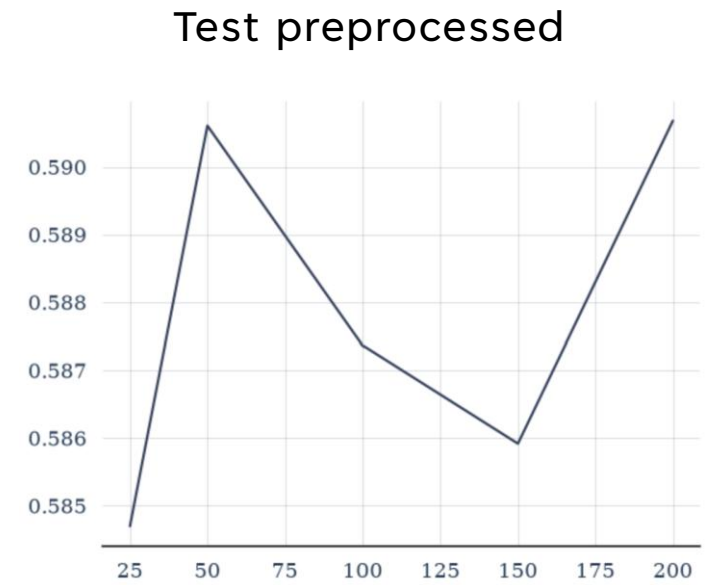
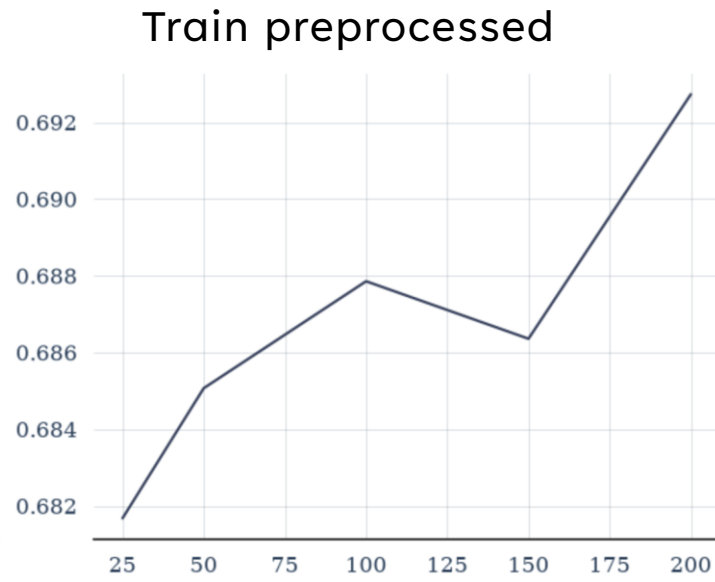
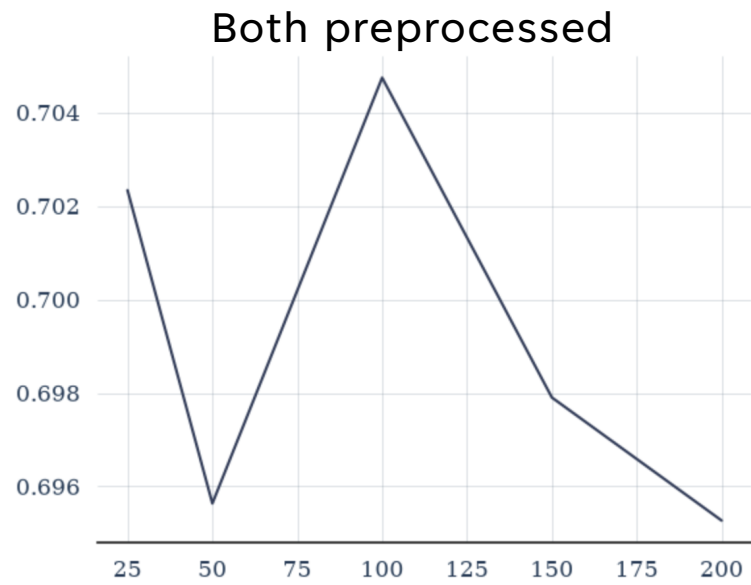


Test preprocessed



WEEK 5

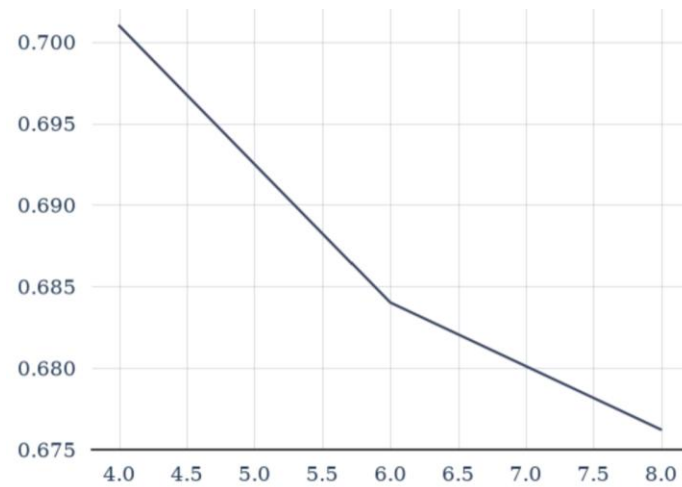
Missing durations:



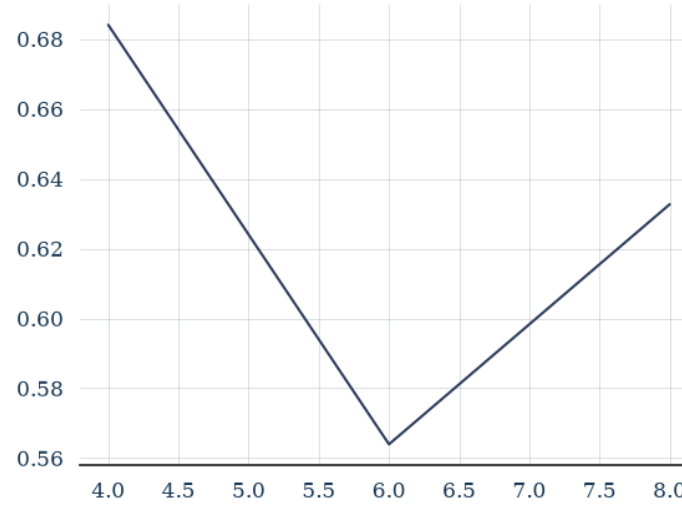
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Missing frequencies:

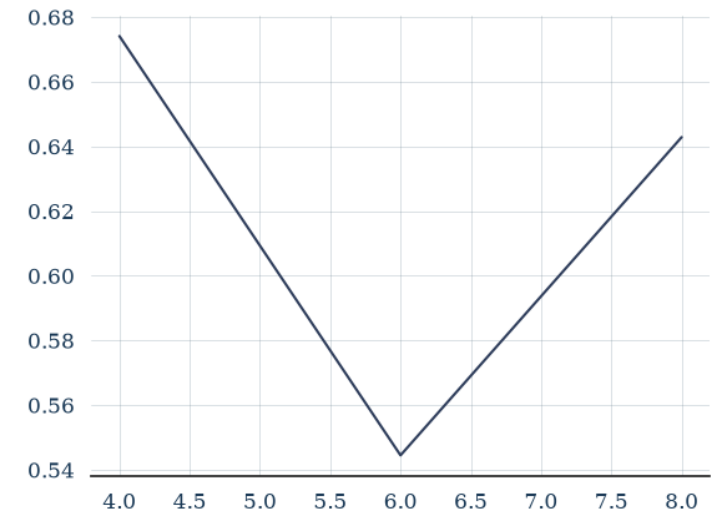
Both preprocessed



Train preprocessed



Test preprocessed



WEEK 5

Baseline 70.5%

	Scaling	Missing durations	Missing frequencies	Missing data	Noise
Both prep.	70.5%	69.9%	68.7%	68.5%	62.7%
Train only	35.6%	68.7%	62.7%	70.1%	45.1%
Test only	30.3%	58.8%	62.1%	39.0%	26.9%